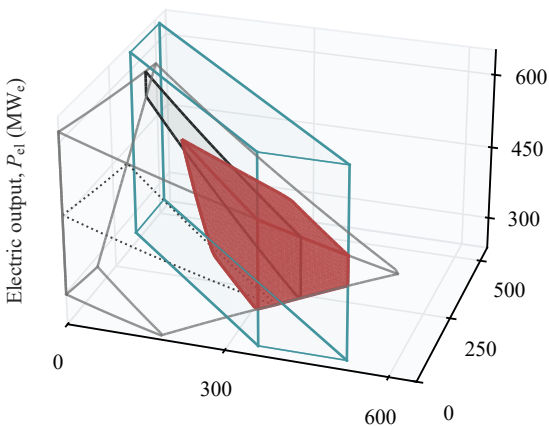
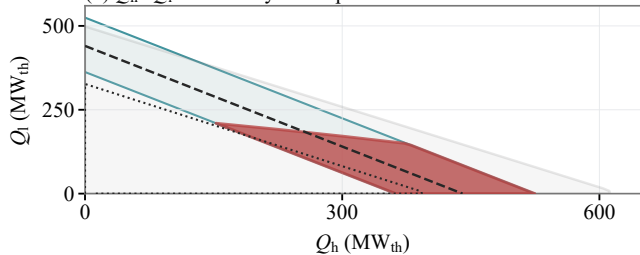


source feasible domain
 coupled admissible domain
 source-load slice, $\lambda=0.75$
 DHN-only envelope
 - - - fixed heat reference

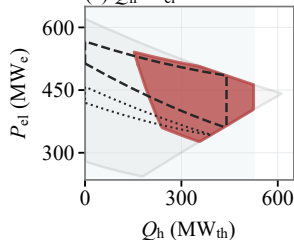
(a) C3: fixed DHN condition, $T_{\text{amb}} = -10\text{ }^{\circ}\text{C}$



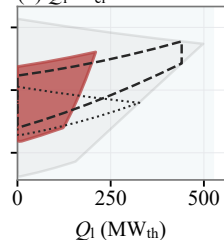
(b) Q_h -- Q_l : DHN-only vs coupled



(c) Q_h -- P_{el}



(d) Q_l -- P_{el}



3D heat-allocation axes: high-grade heat, Q_h , and low-grade heat, Q_l (MW_{th})